Nasr Kasrin (Dr. rer. nat.)

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Technical leader and builder · Architecture & Platforms

Architect, hands-on builder, and engineering leader with 15 years across enterprise platforms (PUMA), a €3.5M EU manufacturing consortium, and a PhD in federated data architecture. I anchor in architecture: designing metadata-driven, AI-native systems that absorb change without breaking the products built on them — and I set the engineering practices that let teams of any seniority build with reliably. Currently co-founder of a production healthcare app (Pflegebericht.com) built with an AI-first methodology I designed. I've seen the pattern from every side: the hard problems are architectural, and the architecture is only as good as the engineering practices behind it.

Work Experience

Co-Founder & Technical Lead **2025 – Present**

Pflegebericht.com (Verhinderungspflege documentation app) *and* dependent AI-adoption advisory

- Co-founded a mobile-first healthcare documentation app for German Verhinderungspflege carers; own product, architecture, and engineering end-to-end (Pflegebericht.com).
- Ran product discovery and planning, shipped v1 to production in 6 weeks (TypeScript/React/Supabase), beta-tested with real carers, then delivered the current stable v2 — full delivery cycle from concept to validated product in one year.
- Designed and executed an AI-assisted engineering methodology (vertical-slice delivery, machine-checkable contracts, AI quality gates) as the app's standard practice — proven on a live production codebase, not a pilot.
- Advised an early-stage healthcare SaaS on modular architecture and AI integration strategy; ran pilot-user focus groups and fed findings directly back into technical design.
- Publish "The Agentic AI Skeleton," a technical series on agentic engineering practice; its harness findings converge with published Databricks and OpenAI research.

Lead Solution Architect **2024 – 2025**

PUMA SE - Global E-commerce Engineering

- Led two engineering teams (10 engineers, 2 leads); ran onboarding and coaching that improved velocity and retention.
- Cut 30% of critical tech debt via a structured effort–value program (CI/CD stabilization, automated testing, reliability), producing a multi-quarter roadmap adopted across teams.
- Guided a cross-team optimization initiative that improved PUMA.com performance by 20% in two quarters.
- Resolved a 90%-order-loss production outage within 24h; established post-mortem and incident practices that cut MTTR by 40%.
- Partnered with product, design, and business stakeholders to shape the technical solution architecture roadmap under growing global traffic.
- Piloted AI/LLM tooling for engineering productivity and code-quality gains; evaluated integration patterns for AI-assisted development across the engineering organization.

Doctoral Candidate & AI Developer **2021 – 2023**

Independent - Germany

- Completed PhD (Dr. rer. nat., Bamberg) on federated inter-organizational data architecture — systems that share and govern data across organizational boundaries without integration. The problem space now addressed by Catena-X and the EU Data Act.
- Ran market assessment and feasibility studies; produced go-to-market scenarios for commercializing the research IP.
- Built an LLM-powered recommendation prototype (muze-ai, Python/LangChain, open source) — hands-on LLM-orchestration work ahead of the current wave.

Research Associate - Team Lead, Architect, Data Solutions Engineer **2015 – 2021**

University of Bamberg - EU Aerospace & Manufacturing Consortium Project (€3.5M, Horizon2020)

- Designed and built an inter-organizational data-sharing system for high-value composites manufacturing - exchanging simulation, test, and measurement data across 8 partners in 5 countries with different tools, conventions, and confidentiality scopes; contributed hands-on to 4 of 6 components.
- Directed delivery of the €3.5M consortium project; owned roadmap, budget, and milestones across multiple European partners.
- Led a cross-functional team of 4 engineers and researchers; ran hiring, onboarding, and mentoring.
- Established team-wide standards for architecture, scalability, security, and reliability.
- Field-tested the federated data-sharing model across three domains (manufacturing, IoT/agriculture, crowd data) - the basis of the later PhD.

R&D Engineer / Team Lead - AI & Mobile **2011 – 2014**

TayaIT - Egypt

- Led a 3-engineer team; delivered AI players that lifted game ratings from 3.0 → 4.5/5.
- Directed iOS app development; achieved 10x early-stage user engagement growth.
- Advised CEO on technical and operational decisions; saved an estimated 10 person-months.

Skills

Architecture & Platform	Solution architecture · Platform architecture (internal development & data platforms) · API design · System design · Product thinking · Technical roadmapping
AI-native engineering	Agentic AI development · LLM evaluation · AI-assisted engineering methodology: harness design, quality gates, machine-checkable contracts · AI/ML system integration
Data architecture	Federated data architecture · (Meta)data strategy & governance · Domain modeling · Semantics
Engineering	Python · TypeScript · React · PostgreSQL · Node.js · REST APIs · Supabase · CI/CD · Reliability engineering · Incident response
Leadership	Cross-functional alignment · Hiring & coaching · Onboarding · Stakeholder management · Leadership without formal authority
Languages	English (fluent) · German (B1/B2)

Education

University of Bamberg - Dr. rer. nat., 2023

Faculty of Information Systems and Applied Computer Sciences. Dissertation: *The Basin Network - A Model for Data Sharing and Exchange.*

German University in Cairo

M.Sc., *Artificial Intelligence & Robotics* B.Sc., *Computer Science & Engineering*